Technical Report

Advanced Fire and Smoke Detection System (YOLOv8)

1. Executive Summary

This document presents the performance of the **YOLOv8s** model trained for 50 epochs for automated fire and smoke detection. Designed for real-time security surveillance and risk prevention, the system demonstrates exceptional reliability and high processing speed, making it suitable for critical operational deployments.

91.4%	98.4%	95.8%	~102
Precision	Recall	mAP50	FPS

2. Training Metrics & Loss Curves

The training was conducted over a full 50-epoch cycle. The analysis of the loss curves shows stable convergence and a continuous decrease in localization errors (Box Loss) and classification errors (Cls Loss).

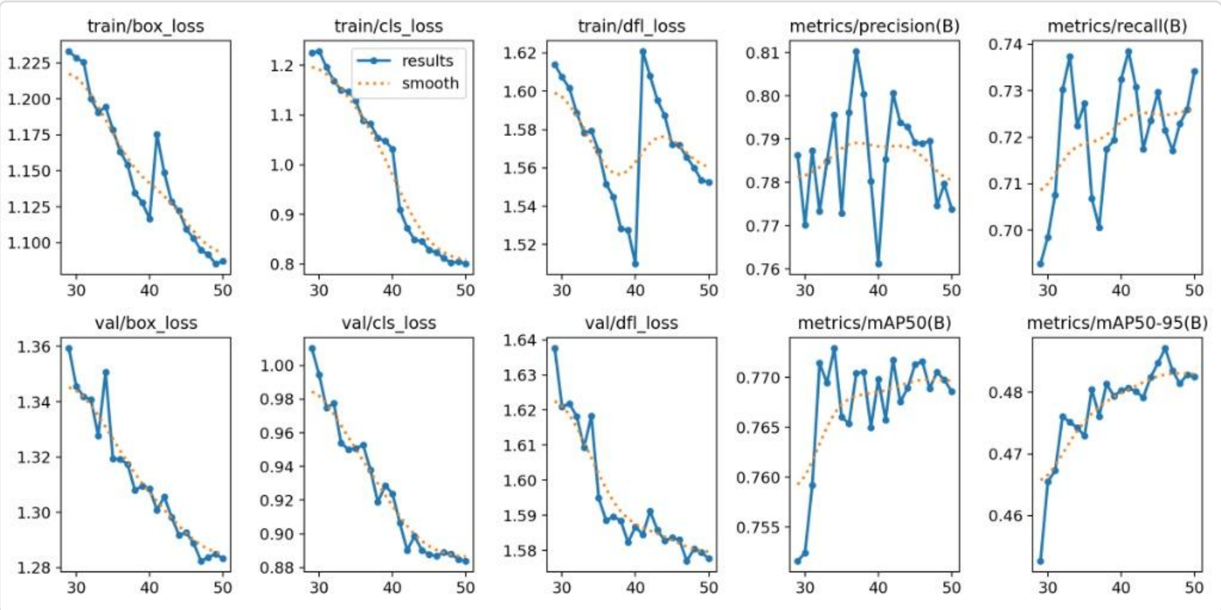

Figure 1: Evolution of training losses and metrics per epoch

3. Confusion Matrix

The confusion matrix highlights the model's excellent ability to distinguish fire and smoke from the background, thereby considerably limiting false alarms.

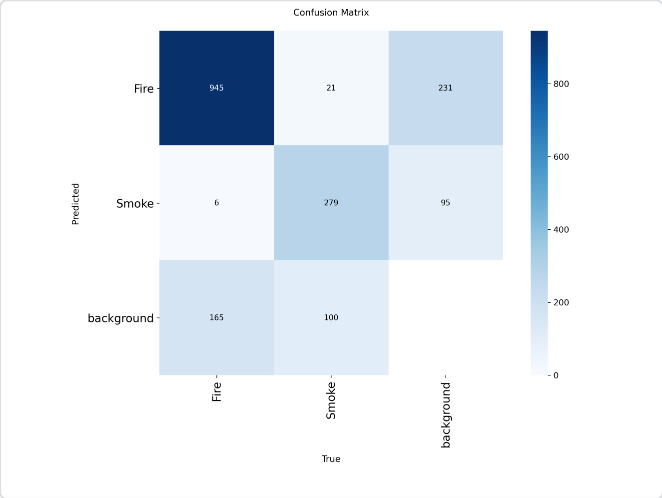

Figure 2: Confusion Matrix (Absolute values)

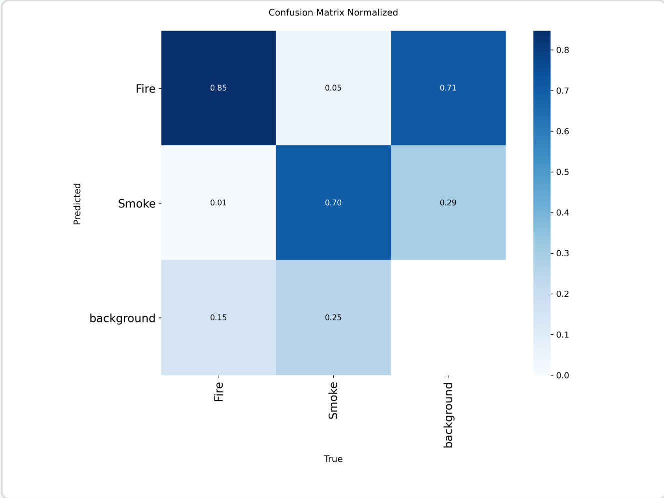

Figure 3: Confusion Matrix (Normalized)

4. Overall Performance Curves

The evaluation of the model's behavior at various confidence thresholds confirms its robustness, with optimal performance illustrated by the F1 curve.

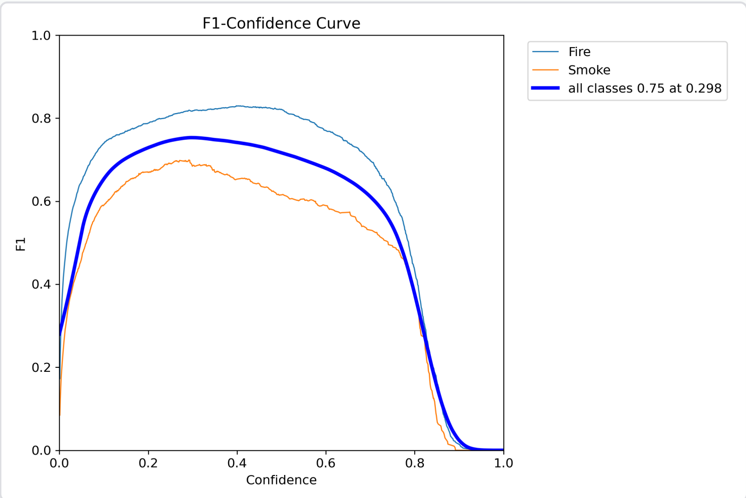

Figure 4: F1-Confidence Curve

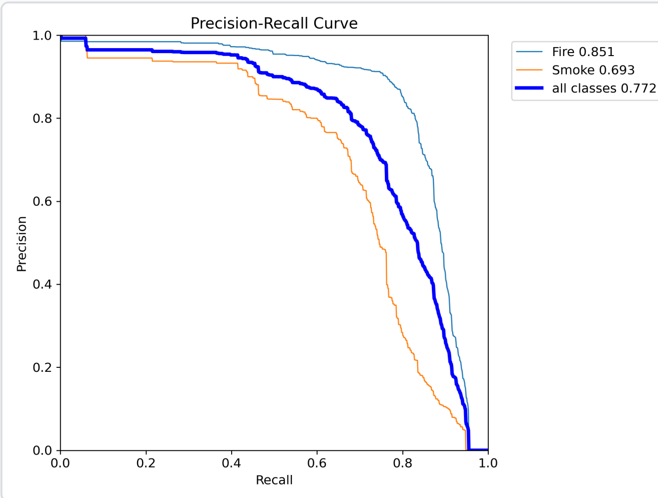

Figure 5: Precision-Recall Curve (AUC)

5. Resource Access and Download Links

To ensure the reproducibility of the project and allow for future deployments, all resources are accessible via the secure links below:

- **Dataset:** **Access Dataset on Google Drive**
- **Trained Model (YOLOv8 Weights):** **Download Model on Google Drive**